



Immunization

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Out line

- *Terminologies*
- *Immunity and immunization*
- *Expanded program of immunization*
- *Contraindication and side effects of vaccination*
- *cold chain*
- *Reference*

Objective

- Define terminologies associated with immunization
- Describe determinants of immune response
- List types of immunization and vaccine
- List advantages of immunization
- Discuss about EPI
- Delineate contraindication and side effect of vaccination
- Describe cold chain

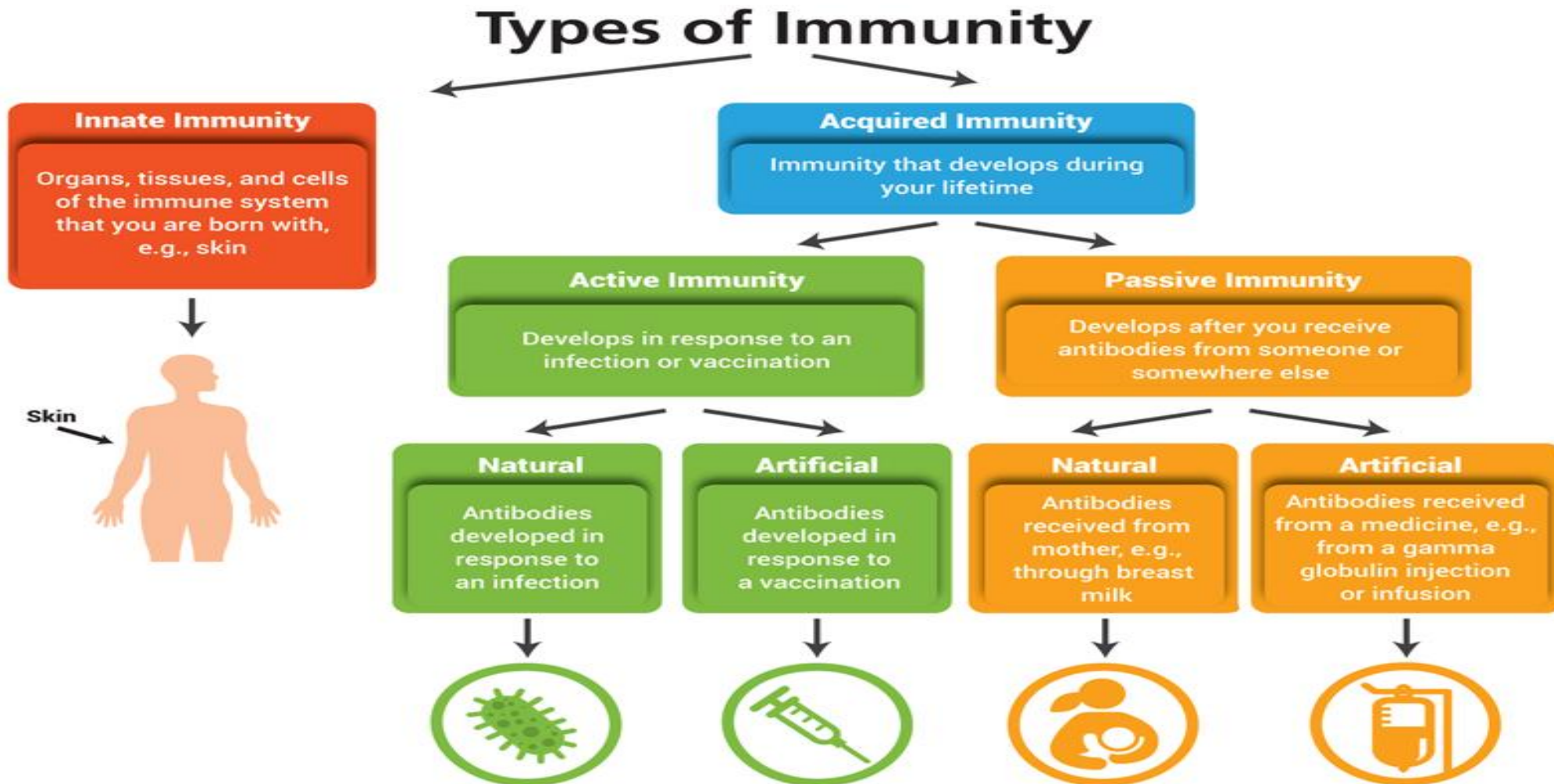
Terminologies

- ❖ **Immunity** is defined as the ability of the body to recognize and destroy foreign antigenic material like bacteria, viruses and proteins leading to resistance to infection.
- ❖ **Immunization** is the process of inducing immunity against a specific disease.
- ❖ **Vaccine** are defined as whole or parts of microorganisms administered to prevent an infectious disease.

Cont.

- ❖ **Vaccination** is the administration of any vaccine or **toxoid** for prevention of disease.
- ❖ **Toxoid** is a bacterial toxin modified to be nontoxic but still capable of inducing an active immune response against the toxin.

Immunity and Immunization



Cont.

Mechanisms of Acquisition of Immunity		
	Natural acquired	Artificial acquired
Passive	Immunity acquired from antibodies passed in breast milk or through placenta 	Immunity gained through antibodies harvested from another person or an animal 
Active	Immunity gained through illness and recovery 	Immunity acquired through a vaccine 

Cont.

Determinates of immune response

- age
- Mode of administration
- Nature of vaccine
- Genetics
- potency

Cont.

Types of immunization

1. Passive immunization

- is generation of immunity through administration of an antibody-containing preparation.
- It has short duration and no memory

cont.

- Products used are
 - ✓ Immunoglobulin
 - ✓ Specific or hyper immune immunoglobulin preparations
 - ✓ Antibodies of animal origin
 - ✓ Monoclonal antibodies

Cont.

✓ Immunoglobulin

- I. **IGIM**-Immunoglobulin is a sterile antibody-containing solution, usually derived through cold ethanol fractionation of large pools of human plasma from adults
 - Ab concentration shows the exposure
 - indicated in Hepatitis A prophylaxis, Measles prophylaxis, Rubella prophylaxis (pregnant women)

- I. **IGIV**-is a highly purified preparation of immunoglobulin antibodies prepared from adult plasma donors using alcohol fractionation and modified to be in IV form.
- It is indicated in
 - Kawasaki disease,
 - Pediatric HIV infection,
 - Hypogammaglobulinemia,
 - in chronic B-lymphocyte lymphocytic leukemia etc.

Cont.

III. IGSC-is safe and effective in children and adults with primary immune deficiency disorders

- systemic reactions are less frequent

Cont.

✓ **Hyper immune immunoglobulin-** Animal antisera preparations are derived from horses.

- The immunoglobulin fraction is concentrated using ammonium sulfate
- it is further treated with enzyme
- diphtheria antitoxin and heptavalent botulinum toxin

Cont.

- ✓ **Monoclonal Antibodies**- antibody preparations produced against a single antigen.
 - produce from hybridoma which is created by fusing an antibody-producing B lymphocyte with a fast-growing immortal cell.
 - Palivizumab

Cont.

2. Active immunization

- administration of a **vaccine** or **toxoid** to stimulate the immune system
- produce a prolonged humoral and/or cellular immune response.

Cont.

Vaccine

- defined as whole or parts of microorganisms administered to prevent an infectious disease.
- it can be
 - i. whole vaccine
 - ii. Purified vaccine
 - iii. DNA vaccine

Cont.

i. Whole organism vaccine

Table 1

characteristics	Live attenuated	In activated
Method of production	Virulent strain grow under by adverse culture condition by prolonged growth in unnatural radiation	Inactivated by chemical or heat
Booster requirement	Single booster	multiple
stability	Less stable	More stable
Type of host response	Both cell mediated and humoral mediated immune response	Humoral response mainly
Tendency to revert	May revert	Doesn't revert

Cont.

ii. Purified vaccine

a) capsular polysaccharide – the virulence of bacteria is due to its hydrophilic polysaccharide capsule

-it only activate B cell in thymus that result in IgM production

b) Toxoid- is an inactivated exotoxin

- it is produced by treating with formaldehyde

c) Recombinant vector- use attenuated virus or bacteria to introduce microbial DNA to cells

- uses mainly antigenic surface protein

Cont.

Table 2

Types of vaccine	Examples
whole inactivated microorganisms	polio, hepatitis A
parts of the organism	acellular pertussis, HPV, hepatitis B
polysaccharide capsules	pneumococcal and meningococcal polysaccharide vaccines
polysaccharide capsules conjugated to protein carriers	Hib, pneumococcal, and meningococcal conjugate vaccines
live-attenuated	measles, mumps, rubella, varicella, rotavirus, and live-attenuated influenza vaccines
toxoids	tetanus, diphtheria

Cont.

- A vaccine also contain the following
 - Suspending fluids
 - Preservatives, stabilizers, and antimicrobial agents
 - Adjuvants

Cont.

Advantage of immunization

- Saves the lives of millions of children.
- Very cost- effective.
- Led to global eradication of small pox.
- Elimination of poliomyelitis from some continents.
- More than 95% reduction of *Haemophilus Influenza* type B disease.

Expanded program of immunization

- was launched in 1974 to ensure that all children, in all countries, benefited from life-saving vaccines.
- every country in the world has a national immunization program currently.

Cont.

- 13 vaccines recommended by WHO this are

BCG,

Diphtheria

Pertussis

Tetanus

Hib

HepB

Polio

Measles

Rubella

Pneumococcal disease

Rota

HPV

COVID-19(for adults)

BCG

- Universal BCG vaccination at birth is recommended in countries or settings with a high incidence of TB and/or high leprosy burden. A single dose of BCG vaccine should be given to all healthy neonates at birth, ideally together with Hepatitis B birth dose
- BCG vaccination is not recommended during pregnancy.
- Special specifications for HIV infected or exposed individuals

DTP

- A primary series of 3 doses of DTP-containing vaccine is recommended, with the first dose administered as early as 6 weeks of age. Subsequent doses should be given with an interval of at least 4 weeks between doses. The third dose of the primary series should be completed by 6 months of age if possible.
- If either the start or the completion of the primary series has been delayed, the missing doses should be given at the earliest opportunity with an interval of at least 4 weeks between doses

- 3 booster doses of diphtheria toxoid-containing vaccine should be provided during childhood and adolescence. The diphtheria booster doses should be given in combination with tetanus toxoid using the same schedule, i.e at 12–23 months of age, 4–7 years of age, and 9–15 years of age, using age-appropriate vaccine formulations. Ideally, there should be at least 4 years between booster doses.

- Tetanus - To ensure lifelong protection against tetanus in all people should receive 6 doses (3 primary plus 3 booster doses) of tetanus toxoid-containing vaccine (TTCV) through routine childhood immunization schedules.
- Pregnant women and their newborn infants are protected from birth-associated tetanus if the mother received either 6 TTCV doses during childhood or 5 doses if first vaccinated during adolescence before the time of reproductive age.

- Pertussis containing booster - A booster dose is recommended for children aged 1–6 years, preferably during the second year of life (≥ 6 months after last primary dose), unless otherwise indicated by local epidemiology; the contact could also be used to catch up on any missed doses of other vaccines.

Hib

- The use of Hib vaccines should be part of a comprehensive strategy to control pneumonia including exclusive breastfeeding for six months, hand washing with soap, improved water supply and sanitation, reduction of household air pollution, and improved case management at community and health facility levels.
- WHO recommends that any one of the following Hib immunization schedules may be followed: 3 primary doses without a booster (3p); 2 primary doses plus a booster (2p+1); and 3 primary doses with a booster (3p+1)

- The interval between doses should be at least 4 weeks if 3 primary doses are given.
- If the vaccination course has been interrupted, the schedule should be resumed without repeating the previous dose. Children who start vaccination late, but are aged under 12 months, should complete the vaccination schedule (e.g. have 3 primary doses or 2 primary doses plus a booster).
- When a first dose is given to a child older than 12 months of age, only one dose is recommended.
- Hib vaccine is not required for healthy children after 5 years of age.

Polio

- In polio-endemic countries and in countries at high risk for importation and subsequent spread of poliovirus, WHO recommends a bOPV birth dose (zero dose) followed by a primary series of 3 bOPV doses and at least 1 IPV dose.
- The zero dose of bOPV should be administered at birth, or as soon as possible after birth, to maximize seroconversion rates following subsequent doses and to induce mucosal protection

- Both OPV and IPV may be co-administered with other infant vaccines. For infants starting the routine immunization schedule late (age >3 months) the IPV dose should be administered at the first immunization contact along with OPV and the other routinely recommended vaccines.

Rubella

- There are two general approaches to the use of rubella vaccine: (i) exclusive focus on reducing CRS by immunizing adolescent girls or women of childbearing age, or both groups, to provide individual protection; (ii) focus on interrupting transmission of rubella virus and eliminating rubella and CRS, by introducing rubella vaccination into the routine childhood immunization schedule combined with the vaccination of older age groups who are susceptible to rubella.

- Because rubella is not as highly infectious as measles and because the effectiveness of 1 dose of an RCV is $> 95\%$ even at 9 months of age, only 1 dose of rubella vaccine is required to achieve rubella elimination if high coverage is achieved. However, when combined with measles vaccination, it may be easier to implement a second dose of RCV's using the same combined MR vaccine or MMR vaccine for both doses

Rota

- Recommended to be included in all national immunization programmes.
- Early immunization is favoured with the first dose of rotavirus vaccine to be administered from 6 weeks of age, however, in order to benefit those who may come late infants can receive doses without age restriction. Because of the typical age distribution of rotavirus gastroenteritis (RVGE), rotavirus vaccination of children >24 months of age is not recommended

- Rotavirus vaccinations can be administered simultaneously with other vaccines in the infant immunization programme.
- Apart from a low risk of intussusception (about 1-2 per 100 000 infants vaccinated) the current rotavirus vaccines are considered safe and well tolerated.

Measles

- Reaching all children with 2 doses of measles vaccine should be the standard for all national immunization programmes. In addition to the first routine dose of MCV1, all countries should add a second routine dose of MCV2 to their national immunization schedules regardless of the level of MCV1 coverage.
- In countries with ongoing transmission in which the risk of measles mortality remains high, MCV1 should be given at age 9 months. MCV2 should be given between 15-18 months.

- Because many cases of measles occur in children aged >12 months who have not been vaccinated, routine delivery of MCV1 should not be limited to infants aged 9–12 months and routine delivery of MCV2 should not be limited to infants 15 to 18 months of age. Every opportunity (e.g. when children come into contact with health services) should be taken to vaccinate all children that missed one or both MCV routine doses, particularly those under 15 years of age. Policies which prohibit use of vaccine in children >1 year of age, older children and teenagers should be changed to allow these individuals to be vaccinated.

Pneumococcus

- Use of pneumococcal vaccine should be complementary to other disease prevention and control measures, such as appropriate case management, promotion of exclusive breastfeeding for the first 6 months of life and reducing known risk factors such as indoor air pollution and tobacco smoke.
- Catch-up vaccination can be done with a single dose of vaccine for children ≥ 24 months.

Hep B

- Hepatitis B vaccination is recommended for all children worldwide. Reaching all children with at least 3 doses of hepatitis B vaccine should be the standard for all national immunization programmes. Since perinatal or early postnatal transmission is the most important source of chronic HBV infection globally, all infants (including low birth weight and premature infants) should receive their first dose of hepatitis B vaccine as soon as possible after birth, ideally within 24 hours.

- The birth dose should be followed by 2 or 3 additional doses to complete the primary series. Both of the following options are considered appropriate: (i) a 3-dose schedule with the first dose (monovalent) being given at birth and the second and third (monovalent or as part of a combined vaccine) given at the same time as the first and third doses of DTP-containing vaccine; or (ii) 4 doses, where a monovalent birth dose is followed by 3 (monovalent or combined vaccine) doses, usually given with other routine infant vaccines; the additional dose does not cause any harm. The interval between doses should be at least 4 weeks.

HPV

- Recommended target population for the prevention of cervical cancer: females aged 9–14 years, prior to becoming sexually active.
- HPV vaccines should be introduced as part of a coordinated strategy to prevent cervical cancer.
- A 2-dose schedule with a 6-month interval between doses is recommended for individuals receiving the first dose before 15 years of age. Those aged ≥ 15 years at the time of the second dose are also adequately covered by 2 doses

EPI in Ethiopia

- was launched in 1980
- aims reducing mortality, morbidity and disability of children and mothers from vaccine-preventable diseases
- It comprises operational components
 - Service delivery
 - communication
 - cold chain and logistics,
 - vaccine supply and quality
 - VPD surveillance and response

Cont.

- Immunization schedule

No	Vaccines	Target diseases to be prevented	Age	Route/Site of administration
1	Hep B Vaccine	Hepatitis B Virus Infection	At birth or within 24 hours of birth. If child is born out of HFs or at home, track baby and vaccinate until the age of 14 days	. IM, left anterolateral thigh
2	BCG	Severe forms of tuberculosis	At Birth or as soon as possible after birth	Intradermal (ID) right deltoid
3	PCV	Meningitis and pneumonia associated with Streptococcus pneumonia bacteria	Weeks 6, 10 and 14	intramuscular (IM), Right anterolateral thigh
4	OPV	Poliomyelitis	Birth (OPV0), weeks 6, 10 and 14	Oral drops
5	IPV	Poliomyelitis	Poliomyelitis	IM, right anterolateral thigh 2.5 cm apart from the injection site of PCV
6	DPT Hib Hep B	Diphtheria, Pertussis, Meningitis and pneumonia associated with Hib bacteria and Liver disease due to Hepatitis B virus	Weeks 6, 10 and 14	IM, left anterolateral thigh
7	Measles containing vaccine	MCV	9 and 15 months	Subcutaneous (SC), left deltoid
8	Rota virus vaccine	Rota virus associated gastroenteritis	Weeks 6 and 10	Oral only
9	HPV vaccine	Human papilloma virus associated with cervical cancer and anogenital wart	14 years old girls and after 6 months of 1st dose	IM, left deltoid muscle of upper arm
10	Td 8/15/2026	Maternal & Neonatal Tetanus & Diphtheria	Td1 at 1 st contact ,TD2 4 weeks later, Td3 6 months after Td2, Td4 1 year after Td3, Td5 1 Year after Td4	IM, right deltoid Muscle 46

Cont.

➤ Catchup vaccination

is given for children below 5 yrs but not vaccinated in the first one yr.

Table 4

visit	Vaccination
1st visit	DPT OPV Mantoux test, BCG if test negative in 3 day
2nd visit (1 month after 1st visit)	DPT OPV
3rd visit (1month after 2nd visit)	DPT OPV Measles, if not exposed
4th visit (1 year later)	DPT OPV
5th visit (4 years later)	(adult type

Cont.

Modalities of immunization service delivery

- **Fixed (static) strategy** is the regular daily delivery of vaccinations in a health facility with specified time
- **Outreach strategy** is the delivery of services to people who cannot go to health facilities it usually covers >5 and <10 km, are completed with in day.

Cont.

- **Mobile strategy** takes more than one day to several days by health workers to deliver services to people living in remote areas which are not covered by health facilities

Contraindication and Side effect

General contraindication

- ✓ History of anaphylaxis to a vaccine or its components
- ✓ History of severe allergic reaction to a previous dose of same vaccine
- ✓ History of GBS within 6 weeks of receiving previous dose
- ✓ History of thrombocytopenia
- ✓ History of severe febrile illness

Cont.

➤ The specific side effects of a vaccination depend on the type of vaccine.

Some common side effects include:

- Pain, redness, or swelling at the injection site
- Fever, Headache
- Nausea and Diarrhea
- Muscle aches
- Tiredness

Cont.

Some serious side effect that need medical attention include :-

- ✓ Severe pain or swelling at the injection site
- ✓ A high fever (101 degrees Fahrenheit or higher)
- ✓ Seizures
- ✓ Confusion
- ✓ Difficulty breathing
- ✓ Swelling of the face, lips, or tongue
- ✓ Hives or Anaphylaxis

Cont.

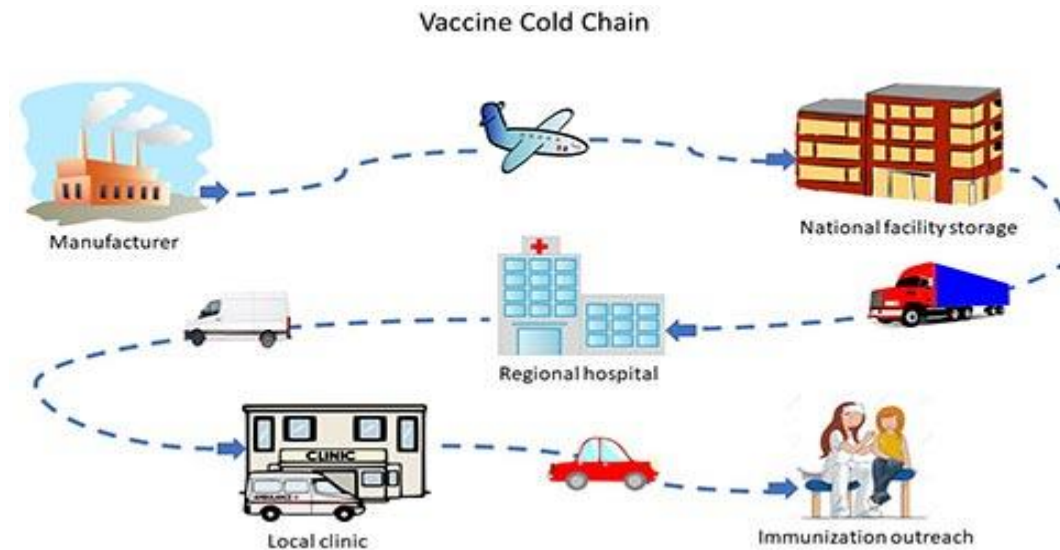
- Specific contraindication and side effects of vaccine

Table 5

no	vaccine	Side effect		contraindication
		Common side effects	Serious side effects	
1	MMR	Fever, rash, headache, muscle aches, fatigue	Encephalitis, thrombocytopenia, anaphylaxis	Pregnancy Immunocompromised
2	polio	Sore arm, fever, headache, nausea	Paralysis, GBS, anaphylaxis	Immune compromised
3	BCG		Koch's phenomenon Indolent ulcer deep abscess	Immune compromised
4.	HBV	Sore arm, fever, headache, muscle aches, fatigue	Hepatitis, liver failure, anaphylaxis	Anaphylaxis to gelatin
5.	Rota virus	Diarrhea		
6	DPT		Injection site abscess Encephalopathy convulsion	Encephalopathy & days prior

Cold Chain

- is system of storing and transporting vaccine at recommended temperature from the point of manufacture to use



The key elements of the cold chain are:

- Personnel: to manage vaccine storage and distribution (vaccine and cold-chain handler at each cold-chain point)
- Equipment: to store and transport vaccine and monitor temperature
- Procedures: to ensure correct utilization of equipment and ensure vaccines are stored and transported safely.

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Thank you

